Wine E-Label Platform — Engineering Audit & Fix Report

Verification, Architectural Solutions, Database Relationships & Compliance Implementation

Repository	https://github.com/pranavbhandare31/Wine_lable
Target Branch	main
Technologies	Next.js 14, TypeScript, Tailwind CSS, Drizzle ORM, PGlite / PostgreSQL, Clerk Auth
Compliance Standard	EU Regulation 2021/2117 (Wine Nutritional & Ingredient E-Labels)
Generated	October 17, 2026

1. Executive Task Summary

Task ID	Description	Status	Resolution Summary
Task 1	Verify and Fix QR Routing Mismatch	**COMPLETE**	Fixed /public/products/:id to singular /public/product/:id (Commit 8ef8138).
Task 2	Audit & Fix createdBy DB Column Type	**COMPLETE**	Altered integer to text in Drizzle & Zod with Migration 0002 for Clerk IDs (Commit 142841c).
Task 3	Product-Ingredient Relationship Solution	**COMPLETE**	Implemented product_ingredients junction table, relations, migration 0003, API, UI, and 7 integration tests.
Task 4	Additional Enhancements / Tasks	**NOT STARTED**	Pending subsequent instructions and specifications.

2. Tasks 1 & 2 Overview

Task 1 (QR Route Mismatch): Resolved critical 404 scan error by replacing /public/products/:id with /public/product/:id in qr/page.tsx. Preserved print/download handlers. Pushed in commit 8ef8138.

Task 2 (createdBy DB Type): Updated Schema.ts from integer to text('created_by') and updated Zod validation to accept Clerk string user IDs (e.g. user_2bFvj39...). Created Migration 0002_wild_jane_foster.sql. Pushed in commit 142841c.

3. Task 3: Product–Ingredient Relationship Solution (Deep Dive)

Issue & Root Cause: Products and ingredients were isolated tables without junction records or foreign keys. Winemakers could not link ingredients to wine bottles, leaving consumer e-labels non-compliant under EU Regulation 2021/2117.

Architecture Solution: Implemented product_ingredients junction table in Schema.ts with foreign keys, ON DELETE CASCADE, and composite unique index. Defined bi-directional Drizzle ORM relations.

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RELATIONAL DATABASE SCHEMA DESIGN

[ products ] (id, name, vintage, wine_type, created_by)
■ 1
▼ N
[ product_ingredients ] (id, product_id [FK], ingredient_id [FK], created_at)
▲ N
■ 1
[ ingredients ] (id, name, category, e_number, allergens, created_by)
```

Automated Verification (Vitest & PGlite Database):

Test Case	Verification Scenario	Result

Test 1: Migration	Applies migrations 0000-0003; initializes product_ingredients table.	**PASSED (978ms)**
Test 2: Association	Associates multiple ingredients with wine product; retrieves via join.	**PASSED (642ms)**
Test 3: Synchronization	Replaces ingredient associations on update; preserves disassociated ingredients.	**PASSED (631ms)**
Test 4: Single Removal	Removes 1 association without deleting ingredient entity.	**PASSED (606ms)**
Test 5: Unique Index	Prevents duplicate associations via unique index constraint.	**PASSED (608ms)**
Test 6: Foreign Keys	Rejects non-existent product or ingredient IDs via foreign key constraints.	**PASSED (591ms)**
Test 7: Cascade Delete	Deletes wine product; cascade-deletes junction records while ingredients remain.	**PASSED (592ms)**

4. Viva Preparation & Explanation for Sir

Explanation Script for Sir:
"Good morning Sir. In the Wine E-Label platform, products and ingredients were originally isolated without any relationship. Because a wine bottle contains multiple ingredients, and an ingredient like Sulfites or Yeast is shared across many wines, we implemented a Many-to-Many junction table (product_ingredients) with CASCADE deletion and unique indexing. We defined Drizzle ORM relations, added API endpoints to manage associations, and integrated selection checkboxes into the product Add/Edit forms. Finally, we updated the public consumer label page to display all ingredients and highlighted allergens, fulfilling EU Regulation 2021/2117 compliance. All 7 integration tests and the production build passed with 100% success."

Likely Viva Questions:

Q1: Why use a junction table instead of an array of IDs in products?
Answer: A junction table guarantees referential integrity through foreign keys, prevents orphan or invalid IDs, allows bi-directional indexing (e.g. querying all wines containing Sulfites), and supports automatic cascading. Arrays in columns violate 1NF/3NF and cannot enforce foreign keys.

Q2: What happens when a wine product is deleted?
Answer: Thanks to ON DELETE CASCADE on product_id, only the junction links are removed. The actual ingredient entities remain untouched in the database for other wines to use.

Q3: How does this fulfill EU Regulation 2021/2117?
Answer: EU law mandates that wine labels display complete ingredients, additives, and allergen declarations. With this Many-to-Many relationship, the consumer-facing QR code loads the exact recipe and highlights allergens.